

Data Dictionary — Visual Product Search Data Model

Every table in `erd.md`, with each attribute's type, key role, and purpose. Types are written as PostgreSQL types (the target database per the tech stack).

BRAND

Attribute	Type	Key	Nullable	Purpose
brand_id	uuid	PK	No	Unique identifier for the brand
name	varchar(120)		No	Display name shown on product pages and used as a search filter

CATEGORY

Attribute	Type	Key	Nullable	Purpose
category_id	uuid	PK	No	Unique identifier for the category
name	varchar(120)		No	Display name (e.g., "Women's Sneakers")
parent_category_id	uuid	FK → CATEGORY.category_id	Yes	Self-referencing link that builds the category hierarchy (root categories have NULL here)

PRODUCT

Attribute	Type	Key	Nullable	Purpose
product_id	uuid	PK	No	Unique identifier for the product
sku	varchar(64)	Unique	No	Business-facing stock keeping unit code
title	varchar(255)		No	Product name shown in search results and used for text search fallback
description	text		Yes	Full product description; source text for any NLP-based re-ranking
brand_id	uuid	FK → BRAND.brand_id	Yes	Links product to its brand
category_id	uuid	FK → CATEGORY.category_id	No	Links product to its leaf category; used to constrain/boost visual search results within a category
base_price	decimal(10,2)		No	List price before variant-level overrides
status	varchar(20)		No	Lifecycle state (active, discontinued, out_of_stock) — filters what's eligible to appear in search results
created_at	timestamp		No	When the product was first listed; used for "new arrivals" and cold-start detection
updated_at	timestamp		No	Last catalog update; drives incremental ETL loads

PRODUCT_VARIANT

Attribute	Type	Key	Nullable	Purpose
variant_id	uuid	PK	No	Unique identifier for a specific color/size combination
product_id	uuid	FK → PRODUCT.product_id	No	Parent product
color	varchar(50)		Yes	Variant attribute; also usable as a visual-search filter/facet
size	varchar(20)		Yes	Variant attribute
price_override	decimal(10,2)		Yes	Overrides base_price for this specific variant, if set
variant_sku	varchar(64)	Unique	No	Variant-level SKU, distinct from the parent product SKU

Why variants exist as their own table: color and size directly change what a product photo looks like, so visual search needs to match against the correct variant's images — not just the parent product.

PRODUCT_IMAGE

Attribute	Type	Key	Nullable	Purpose
image_id	uuid	PK	No	Unique identifier for the image
variant_id	uuid	FK → PRODUCT_VARIANT.variant_id	No	Which variant this image depicts
image_url	varchar(500)		No	CDN path to the image file
image_type	varchar(20)		No	<code>catalog</code> , <code>marketing</code> , or <code>lifestyle</code> — lets the pipeline exclude staged marketing photos from model training (per Marketing stakeholder feedback, Task 1)
width	int		Yes	Pixel width, used for pre-resize/normalization before inference
height	int		Yes	Pixel height
is_primary	boolean		No	Marks the default image shown in search results/listings
uploaded_at	timestamp		No	When the image was added

IMAGE_EMBEDDING

Attribute	Type	Key	Nullable	Purpose
embedding_id	uuid	PK	No	Unique identifier for the embedding record
image_id	uuid	FK → PRODUCT_IMAGE.image_id	No	Which image this embedding was generated from
embedding_vector	vector(512)		No	The actual feature vector output by the TensorFlow model (pgvector type); this is what visual similarity search matches against
model_version	varchar(30)		No	Which model produced this embedding — critical for re-indexing when the model is retrained, since old and new embeddings aren't comparable
created_at	timestamp		No	When the embedding was generated

This is the table that makes "visual" search possible. Everything else in the schema supports catalog and business logic; this table is the actual input to the nearest-neighbor lookup that finds visually similar products.

USER

Attribute	Type	Key	Nullable	Purpose
user_id	uuid	PK	No	Unique identifier for the user
email	varchar(255)	Unique	No	Login/contact identifier
created_at	timestamp		No	Account creation date; used for cohort analysis and new-user cold-start handling

(Additional profile/preference fields intentionally omitted here — see Task 1 report, Section 2, "User Profiles & Preferences" source; that's a separate service and out of scope for the visual-search-specific model.)

USER_INTERACTION

Attribute	Type	Key	Nullable	Purpose
interaction_id	uuid	PK	No	Unique identifier for the event
user_id	uuid	FK → USER.user_id	No	Who performed the action
product_id	uuid	FK → PRODUCT.product_id	No	Which product was acted on
interaction_type	varchar(30)		No	<code>view</code> , <code>click</code> , <code>add_to_cart</code> , <code>wishlist</code> , etc. — the label used to train ranking/relevance models
source	varchar(20)		No	<code>text_search</code> or <code>visual_search</code> — critical for measuring whether visual search actually improves engagement vs. traditional search (ties directly to the \$2.5M business problem)
occurred_at	timestamp		No	Event timestamp; streamed from Kafka

SEARCH_QUERY

Attribute	Type	Key	Nullable	Purpose
query_id	uuid	PK	No	Unique identifier for the search event
user_id	uuid	FK → USER.user_id	Yes	Who searched (nullable to support anonymous/guest search)
query_type	varchar(10)		No	<code>text</code> or <code>image</code> — distinguishes traditional from visual search
query_text	varchar(500)		Yes	Raw text query, if query_type = text
query_image_id	uuid	FK → PRODUCT_IMAGE.image_id or external upload ref	Yes	The uploaded/reference image, if query_type = image
results_count	int		No	How many results were returned — used to detect zero-result queries, a key visual-search quality signal
clicked_product_id	uuid	FK → PRODUCT.product_id	Yes	First product clicked from these results, if any — used to measure search relevance
timestamp	timestamp		No	When the query was submitted

REVIEW

Attribute	Type	Key	Nullable	Purpose
review_id	uuid	PK	No	Unique identifier for the review
product_id	uuid	FK → PRODUCT.product_id	No	Product being reviewed
user_id	uuid	FK → USER.user_id	No	Author of the review
rating	int		No	1-5 star rating
review_text	text		Yes	Free-text review content
created_at	timestamp		No	When the review was posted

REVIEW_IMAGE

Attribute	Type	Key	Nullable	Purpose
review_image_id	uuid	PK	No	Unique identifier for the image
review_id	uuid	FK → REVIEW.review_id	No	Parent review
image_url	varchar(500)		No	CDN path to the user-uploaded image

Why this matters for visual search: review images are real-world, user-taken photos — visually very different from staged catalog photography. They're a valuable secondary training signal for making the visual search model robust to photos users actually upload (per Marketing's feedback in Task 1: don't overfit purely to staged imagery).

ORDER_

(Named `ORDER_` to avoid clashing with the SQL reserved word `ORDER`.)

Attribute	Type	Key	Nullable	Purpose
order_id	uuid	PK	No	Unique identifier for the order
user_id	uuid	FK → USER.user_id	No	Who placed the order
order_date	timestamp		No	When the order was placed
total_amount	decimal(10,2)		No	Total order value — used to tie visual search sessions back to actual revenue
status	varchar(20)		No	Order lifecycle state (placed, shipped, returned, refunded)

ORDER_ITEM

Attribute	Type	Key	Nullable	Purpose
order_item_id	uuid	PK	No	Unique identifier for the line item
order_id	uuid	FK → ORDER_order_id	No	Parent order
variant_id	uuid	FK → PRODUCT_VARIANT.variant_id	No	Which specific variant was purchased
quantity	int		No	Units purchased
unit_price	decimal(10,2)		No	Price at time of purchase (kept separate from current price for historical accuracy)

WAREHOUSE

Attribute	Type	Key	Nullable	Purpose
warehouse_id	uuid	PK	No	Unique identifier for the fulfillment center
name	varchar(120)		No	Display name
location	varchar(255)		No	Physical location, used for "in stock near me" logic

INVENTORY

Attribute	Type	Key	Nullable	Purpose
inventory_id	uuid	PK	No	Unique identifier for the stock record
variant_id	uuid	FK → PRODUCT_VARIANT.variant_id	No	Which variant this stock count applies to
warehouse_id	uuid	FK → WAREHOUSE.warehouse_id	No	Which warehouse holds this stock
stock_qty	int		No	Current available quantity — used to filter/de-rank out-of-stock items in visual search results
updated_at	timestamp		No	Last sync time from the ERP/WMS system

Notes on completeness

This dictionary intentionally covers only the entities needed to **serve visual product search end to end**: catalog structure (Brand/Category/Product/Variant), the imagery + AI layer (Image/Embedding), behavioral signal (Interaction/SearchQuery), trust signal (Review/ReviewImage), and the commercial layer that justifies the feature (Order/OrderItem/Inventory/Warehouse). Broader account/identity data (addresses, payment methods, notification preferences) lives in other services per the Task 1 inventory and is out of scope here to keep the model focused on what visual search actually needs.